Global Olympic Sports Performance Analysis

Medal Achievements, Regional Efficiency and Investment using Tableau

Final Project Report

Submitted as part of the SkillWallet Internship Program
(Data Analytics with Tableau)

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1. Introduction

1.1 Project Overview

The Olympic Games are a global platform where nations showcase their sporting excellence, yet analysing what drives Olympic success is difficult because of the large volume of historical data from 1992 to 2026. This project builds an interactive Tableau solution on the Kaggle dataset "Olympic Games Performance &

Medal History 1992–26", which contains four related files: athletes.csv (8,000 rows), medal_results.csv (910 rows), games_summary.csv (18 rows) and alltime_country_rankings.csv (60 rows).

The four files were related in Tableau and prepared with seven calculated fields and a medal-type parameter. Twelve unique visualizations and four KPI cards were combined into two responsive dashboards — the Global
Sports Performance Analytics Dashboard and the Olympic Performance and Regional Efficiency Dashboard — and an eight-scene story. The workbook was published to Tableau Public and embedded in a Flask web application.

1.2 Purpose

The purpose of this project is to help three user groups make data-backed decisions:

- **National Olympic Committee strategists**, who need to know which sport categories give the best medal return per athlete and how their country compares with leaders such as the United States (735 medals) and China (696 medals).
- **Sports investment analysts**, who need to see how investment levels relate to medal performance and which regions are most efficient.
- **Sports journalists and data storytellers**, who need clear, data-backed narratives about host-nation advantage, medal trends and regional dominance.

2. Ideation Phase

2.1 Problem Statement

Two customer problem statements were defined to frame the project from the perspective of its primary users.

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a performance strategist at a National Olympic Committee	decide which sport categories and disciplines to invest in for the next Olympic cycle	I cannot see which categories give the best medal return per athlete	medal data for 60 countries over 18 Games (1992–2026) is spread across separate files and is not summarised visually	unsure and worried about wasting limited funding
PS-2	a sports investment analyst at an international funding agency	judge whether a country's sports investment actually produces medals	I have no clear view comparing investment levels with medal efficiency by region	investment, efficiency and medal figures sit in different tables and must be crossreferenced by hand	frustrated by slow, manual analysis before grant decisions

2.2 Empathy Map Canvas

An empathy map was built for "Priya", a persona representing a National Olympic Committee performance strategist, to capture what she thinks, feels, sees, hears, says and does. Key pain points were the difficulty of comparing sports by medal return, the lack of a single view of regional efficiency, and being overwhelmed by multi-file data. The gains the solution targets are a clear ranking of high-yield sport categories, an instant view of efficiency and investment by region, and faster evidence-based funding decisions.

2.3 Brainstorming

Brainstorming started from the selected problem statement: "How might we help Olympic committees, investors and sports journalists understand what drives Olympic medal success between 1992 and 2026?" Ideas were grouped into four themes — Medal Performance, Trends & Hosts, Regional & Investment, and Usability & Web — and placed on an importance-versus-feasibility grid.

Idea	Importance	Feasibility
Bar chart: Total Medals by Country	High	High — built (Dashboard 1)
Line chart: Trend of Medals Over the Years	High	High — built (Dashboard 1)
Map: Total Medal Achievements Across Nations	High	High — built (Dashboard 1)
Bar chart: Avg Medals per Athlete by Sport Category	High	High — built (Dashboard 1)
Pie: Avg Medal Efficiency by Region	High	High — built (Dashboard 2)
Packed bubbles: Sports Investment vs Medals	High	High — built (Dashboard 2)
Bar chart: Medal Totals Across Olympic Hosts	High	High — built (Dashboard 2)
Donut: Medal Share by Region / Radial gauge	Medium	High — built (Dashboard 2)
Treemap and connected dot plot	Medium	High — built (Story)
Predictive model for future medal counts	Medium	Low — deferred (future scope)

3. Requirement Analysis

3.1 Customer Journey Map

The journey map traces a strategist or analyst from first accessing the website, through reviewing KPIs, analysing visualizations and reviewing the story, to making a funding or strategy decision.

Stage	Access Dashboard	Review KPIs	Analyze Visualizations	Review Story	Make Decisions
Experience steps	Opens the Flask website or Tableau Public link to view the Olympic analysis.	Scans KPI cards: 6,025 medals, 60 nations, 8,000 athletes, 18 Games.	Uses Year, Region, Country, Sport Category and Host Country filters and the medal-type parameter.	Walks through the 8-scene guided story.	Applies insights to sport funding, grant allocation or media stories.
Interactions	Website home page / Tableau Public.	KPI tiles and efficiency gauge.	Interactive filters, parameter and colourcoded charts.	Story navigator with captions and descriptions.	Team discussions, shared links and screenshots.
Goals & motivations	Help me get a quick overview of Olympic performance.	Help me understand the headline numbers.	Help me find patterns worth acting on.	Help me see the big picture, not isolated charts.	Help me act with confidence.
Positive moments	Clean navy-and-gold layout loads quickly.	KPIs make overall scale easy to grasp.	Filters reveal host advantage and regional efficiency.	Story turns numbers into a clear narrative.	Decision feels backed by real data.

Negative moments	Unsure where to start among many charts.	Efficiency % needs context to interpret.	Many filter combinations can feel overwhelming.	Story is static unless paired with live dashboards.	Wants more recent or predictive data.
Areas of opportunity	Add a short 'how to read' note.	Add benchmark context to KPIs.	Offer default filter presets.	Refresh story after each Games.	Add exportable summary and forecasts.

3.2 Solution Requirement

Functional requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection & Extraction	Download the 4 CSV files from Kaggle; connect them to Tableau; create relationships on noc_code and year + season
FR-2	Data Preparation	Check null values and duplicates; set data types; convert Year to a discrete dimension; build calculated fields and a parameter
FR-3	Data Visualization	Build 12 unique visualizations (bar, line, map, pie, donut, treemap, packed bubbles, connected dot plot, radial gauge) and 4 KPI cards
FR-4	Dashboard Creation	Build 2 responsive dashboards; add Year, Sport Category, Country, Region and Host Country filters and the Select Medal parameter
FR-5	Storyboard Creation	Build an 8-scene story with captions and descriptions
FR-6	Performance Testing	Document data loaded, filters, calculated fields and number of visualizations
FR-7	Web Integration	Publish to Tableau Public; embed dashboards and story in a Flask web application
FR-8	Documentation & Demonstration	Record a 5–7 minute explanation video; write step-by-step project documentation

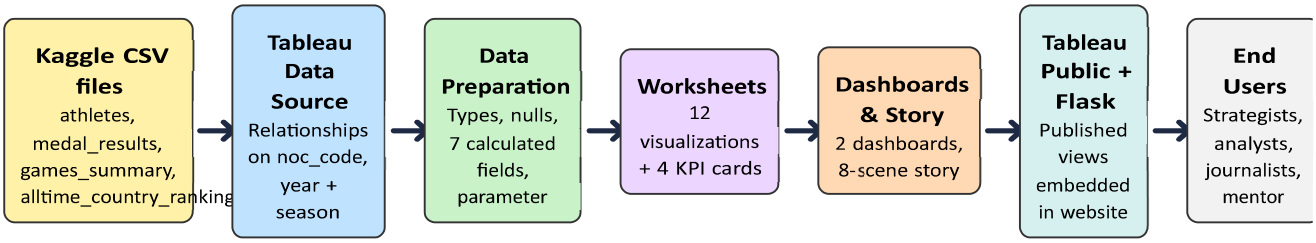
Non-functional requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Consistent navy-and-gold design, clear titles, labels and tooltips so nontechnical users can read insights without training.
NFR-2	Security	Only public, non-personal data is used; workbook and code shared through Tableau Public and GitHub links.
NFR-3	Reliability	KPI values (6,025 medals, 60 nations, 8,000 athletes, 18 Games) always match the source data.
NFR-4	Performance	Dashboards load and respond to filters within a few seconds for ~9,000 rows.
NFR-5	Availability	Tableau Public views and the GitHub repository remain accessible for mentor review.

NFR-6	Scalability	New Games, countries or sheets can be added without redesigning the dashboards.

3.3 Data Flow Diagram

The diagram below shows how data moves from the raw CSV files to the end users.

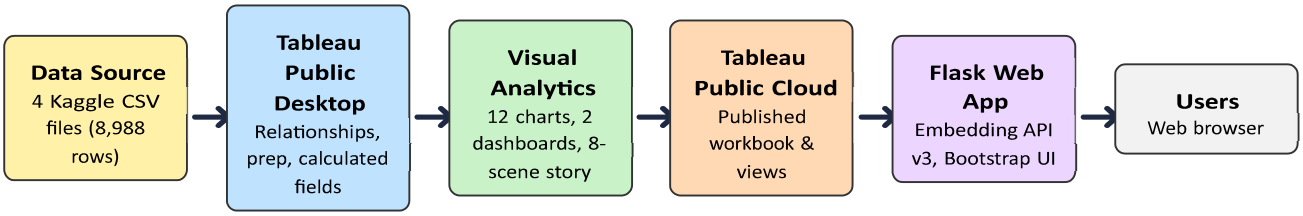


Selected user stories driving this flow are summarized below.

User Story No.	Epic	User Story	Priority
USN-1	Data Collection & Extraction	Connect and relate the 4 Olympic CSV files in Tableau	High
USN-2	Data Preparation	Verify data types, nulls and duplicates	High
USN-3	Data Preparation	Create calculated fields and the Select Medal parameter	High
USN-4	Data Visualization	Create 12 charts and 4 KPI cards	High
USN-5	Dashboard	Combine charts into 2 interactive dashboards	High
USN-6	Storyboard	Create an 8-scene guided story	Medium
USN-7	Performance Testing	Document data volume, filters and calculations	Medium
USN-8	Web Integration	Embed dashboards and story in a Flask website	High
USN-9	Documentation & Demo	Record demo video and write documentation	High
USN-10	Project Review	Mentor reviews published views, code and video	High

3.4 Technology Stack

The architecture flows from the raw CSV files, through Tableau Public Desktop for preparation and visualization, to Tableau Public Cloud, and finally into a Flask web application viewed in a browser.



S.No	Component	Description	Technology
1.	User Interface	How users view the dashboards and story	Flask web pages (HTML, CSS, Bootstrap 5), Tableau Public web view, Web browser

2.	Application Logic1	Relates and prepares the 4 CSV files (types, nulls, discrete Year)	Tableau Public Desktop (Data Source)
3.	Application Logic2	Builds visualizations, calculated fields, parameter, dashboards and story	Tableau Public Desktop
4.	Application Logic3	Serves web pages and embeds published views	Python Flask, Tableau Embedding API v3
5.	Database	No traditional database; data comes from CSV files	Not Applicable for this project
6.	Cloud Database	Not Applicable for this project	Not Applicable for this project
7.	File Storage	Raw CSV files, Tableau workbook and Flask source code	Local File System (CSV, .twbx), GitHub
8.	External API-1	Embeds Tableau Public views into web pages	Tableau Embedding API v3 (JavaScript)
9.	External API-2	Not Applicable for this project	Not Applicable for this project
10.	Machine Learning Model	Not Applicable for this project	Not Applicable for this project
11.	Infrastructure (Server / Cloud)	Local development; dashboards hosted on Tableau Public; website run locally with Flask; code on GitHub	Local Machine, Tableau Public, Flask dev server, GitHub

4. Project Design

4.1 Problem Solution Fit

Customer segments: National Olympic Committee strategists, sports investment analysts, and sports journalists.

Jobs-to-be-done: find which countries, regions and sport categories perform best; judge whether investment and hosting improve results; and tell clear stories about 34 years of Olympic history.

Available alternatives — official medal tables, news articles, raw CSV files and one-off spreadsheets — show medal counts but not efficiency, investment or host advantage together. The root cause is that data for 60 countries and 18 Games is split across four files with no single visual tool combining them.

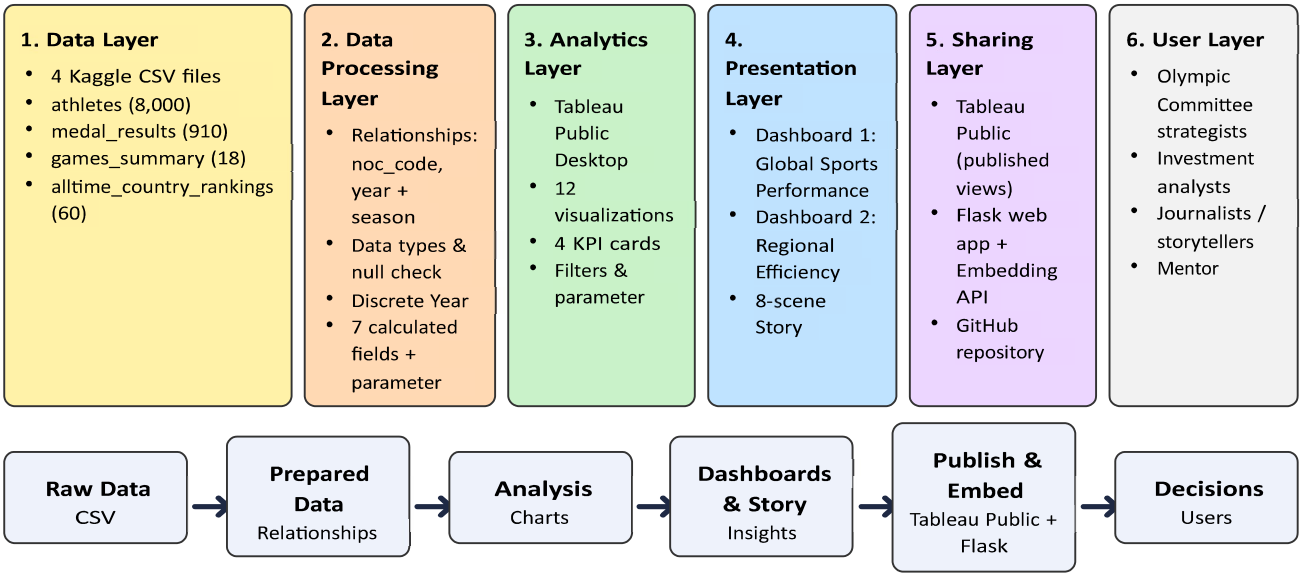
The proposed solution — two interactive dashboards and an eight-scene story published on Tableau Public and embedded in a Flask website — moves users from being overwhelmed by scattered data to being confident in their funding and storytelling decisions. **4.2 Proposed Solution**

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Olympic committees, investors and journalists struggle to understand what drives medal success. Data from 1992 to 2026 covering 60 countries, 18 Games and 8,000 athletes is spread across four files, making it hard to compare countries, regions, sport categories, investment and host advantage.

2.	Idea / Solution description	An interactive Tableau solution built on four related CSV files: a Global Sports Performance Analytics Dashboard (KPIs, world map, medal trend, sport categories, medals by country with a medal-type parameter) and an Olympic Performance & Regional Efficiency Dashboard (efficiency gauge, regional efficiency, investment bubbles, host-nation totals, medal share), plus an 8-scene story — all embedded in a Flask website.
3.	Novelty / Uniqueness	Combines medal counts with efficiency, investment level and host status in one place; uses a parameter to switch medal types; and delivers the analysis both as a guided story and inside a custom website rather than as isolated charts.
4.	Social Impact / Customer Satisfaction	Helps committees fund the right sports, helps funding agencies spot efficient but underfunded regions (e.g. South America at 49.6% efficiency), and gives journalists databacked stories.
5.	Business Model (Revenue Model)	Free on Tableau Public as an internship project. A real-world extension could offer a premium version with live data, forecasts and country-level reports for federations and media.
6.	Scalability of the Solution	New Games, countries or sports can be added to the CSV files and refreshed without redesign; the Flask site can host more pages or be deployed to the cloud.

4.3 Solution Architecture

The solution is organized into six layers: a Data Layer (4 Kaggle CSV files), a Data Processing Layer (relationships, data types, calculated fields), an Analytics Layer (Tableau Public Desktop, visualizations, filters), a Presentation Layer (two dashboards and the story), a Sharing Layer (Tableau Public, Flask web app, GitHub) and a User Layer (strategists, analysts, journalists, mentor).



5. Project Planning & Scheduling

5.1 Project Planning

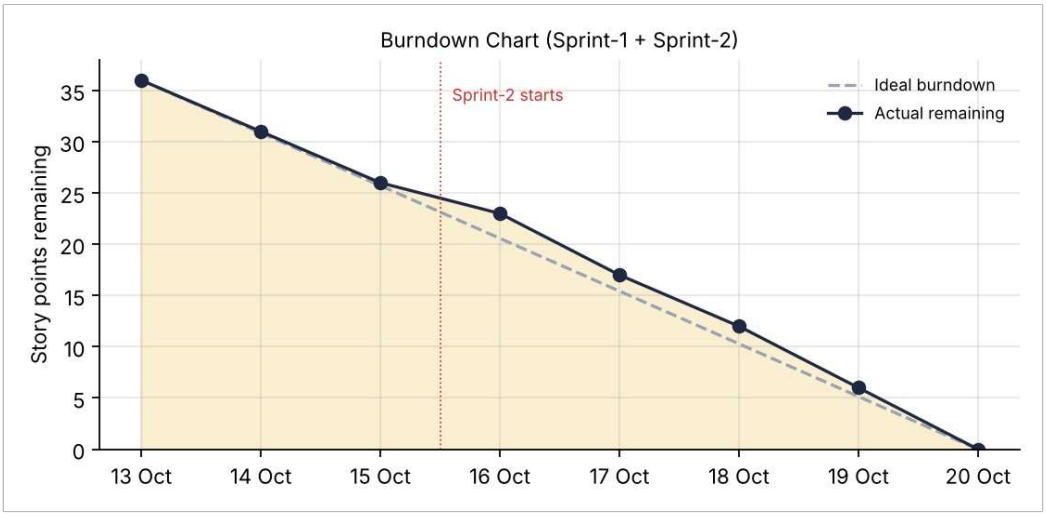
Work was organized into two sprints: data collection and preparation, followed by visualization, dashboards, story, testing, web integration and documentation.

Sprint	Epic	User Story	Points	Priority	Owner
1	Data Collection & Extraction	As a user, I can download the 4 CSV files and connect them to Tableau	2	High	Manas Geedkar
1		As a user, I can relate the tables on noc_code and year + season	3	High	Manas Geedkar
1	Data Preparation	As a user, I can verify data types, nulls and set Year as discrete	2	High	Manas Geedkar
1		As a user, I can create calculated fields and the Select Medal parameter	5	Medium	Manas Geedkar
2	Data Visualization	As a user, I can view bar, line, map and pie charts of medal performance	5	High	Manas Geedkar
2		As a user, I can view donut, treemap, bubbles, dot plot, gauge and KPI cards	5	High	Manas Geedkar
2	Dashboard	As a user, I can explore 2 responsive dashboards with filters	5	High	Manas Geedkar
2	Story	As a user, I can follow an 8-scene guided story	3	Medium	Manas Geedkar
2	Performance Testing	As a user, I can review documented data volume, filters and calculations	1	Medium	Manas Geedkar
2	Web Integration	As a user, I can open the dashboards and story in a Flask website	3	High	Manas Geedkar
2	Documentation & Demo	As a user, I can watch a demo video and read the documentation	2	High	Manas Geedkar

Sprint tracker:

Sprint	Total Points	Duration	Start Date	Planned End	Completed	Actual Release
Sprint-1	12	3 Days	13 Oct 2026	15 Oct 2026	12	15 Oct 2026
Sprint-2	24	5 Days	16 Oct 2026	20 Oct 2026	24	20 Oct 2026

Average velocity (AV) = 36 story points ÷ 8 days = 4.5 story points per day. Both sprints were completed on schedule.


Burndown Chart (Sprint-1 + Sprint-2)
Story points remaining
35
30
25
20
15
10
5
0
Sprint-2 starts
Ideal burndown
Actual remaining
13 Oct
14 Oct
15 Oct
16 Oct
17 Oct
18 Oct
19 Oct
20 Oct

6. Functional and Performance Testing

6.1 Performance Testing

S.No.	Parameter	Screenshot/values
1.	Data Rendered	4 CSV files with 8,988 rows rendered successfully in Tableau: athletes.csv (8,000 rows × 26 cols), medal_results.csv (910 × 19), games_summary.csv (18 × 10), alltime_country_rankings.csv (60 × 13). Tables kept as flat CSVs and combined through relationships; Tableau's in-memory engine handles all aggregations.
2.	Data Preprocessing	Relationships on noc_code and year + season; nulls and duplicates checked; data types validated; Year converted to discrete; Host Nation converted to dimension.
3.	Utilization of Filters	Dashboard 1: Year (range slider – timeline filter), Sport Category, Country, and Select Medal parameter. Dashboard 2: Region, Host Country, Year. Sheet-level: Host Nation = 1 (hosts chart), Top 15 country filters, Overall Rank 1–15 (dot plot). Filters set to apply to all worksheets using the data source.
4.	Calculation fields Used	7 calculated fields + 1 parameter: Efficiency Ratio, Remaining, Medals per Athlete, Gold Share %, Host Advantage, Olympic Edition, Selected Medals; parameter Select Medal (Total / Gold / Silver / Bronze).
5,	Dashboard design	No of Visualizations / Graphs – 2 dashboards. Dashboard 1: 4 KPIs + map, line, 2 bar charts. Dashboard 2: 3 KPIs + radial gauge, pie, packed bubbles, bar, donut. Automatic (responsive) size, navy background with gold title bands.
6.	Story Design	No of Visualizations / Graphs – 8 story points (Overview dashboard, Medal trend, Top countries, Host advantage, Investment bubbles, Summer vs Winter dot plot, Regional efficiency dashboard, Treemap).
7.	No of Visualizations / Graphs	12 unique visualizations + 4 KPI cards: horizontal bars (3), line, filled map, donut, treemap, packed bubbles, connected dot plot, radial gauge, pie.

7. Results

7.1 Output Screenshots

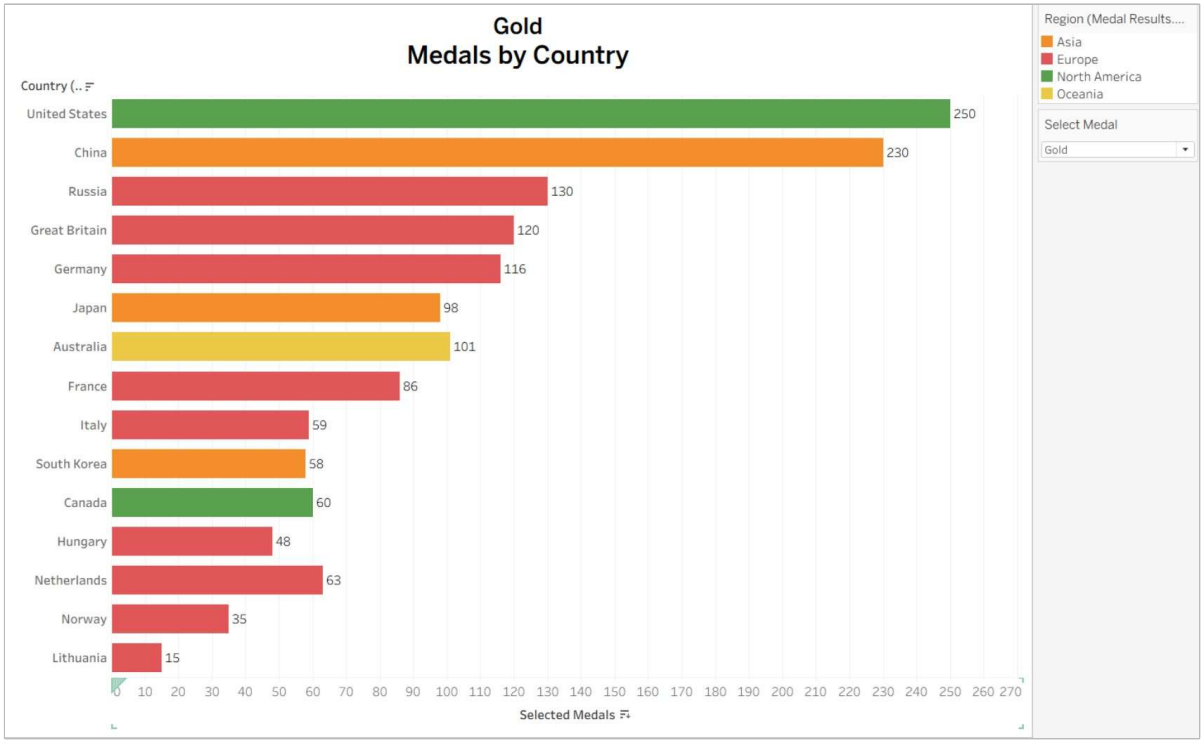


Figure 1: Total Medals by Country (Horizontal Bar) – Select Medal parameter set to Gold

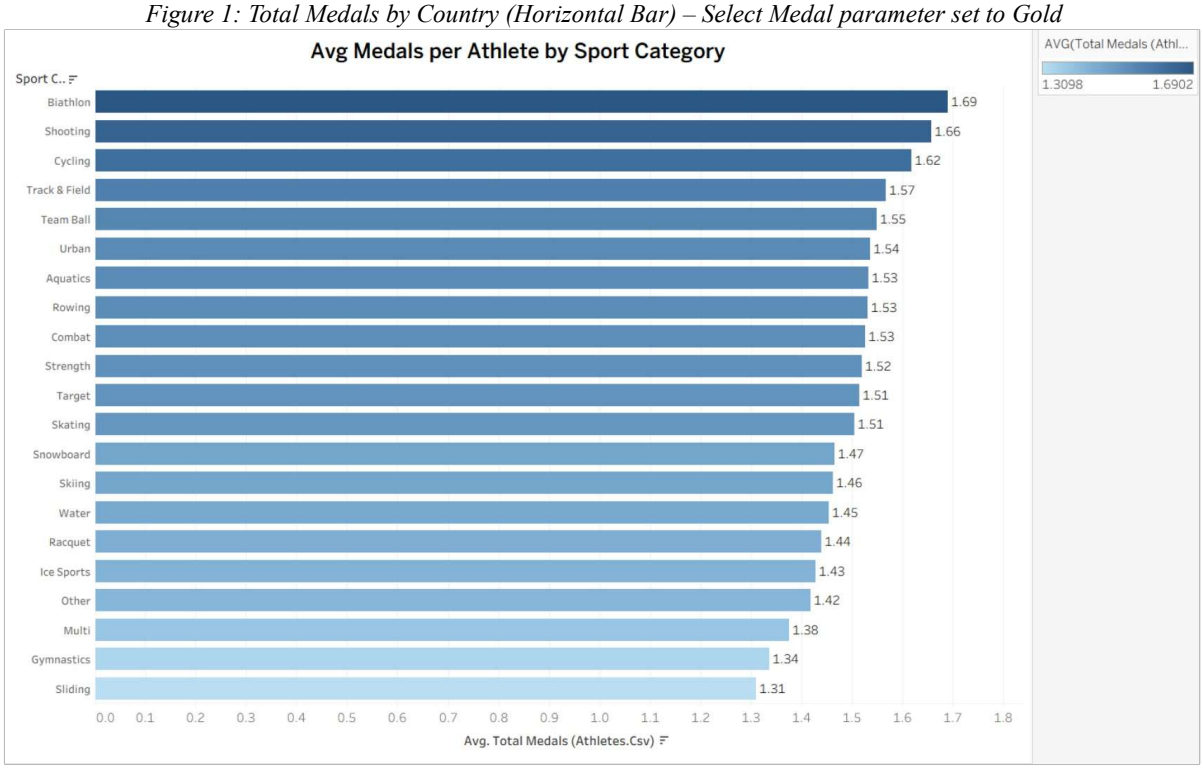


Figure 2: Avg Medals per Athlete by Sport Category (Horizontal Bar)

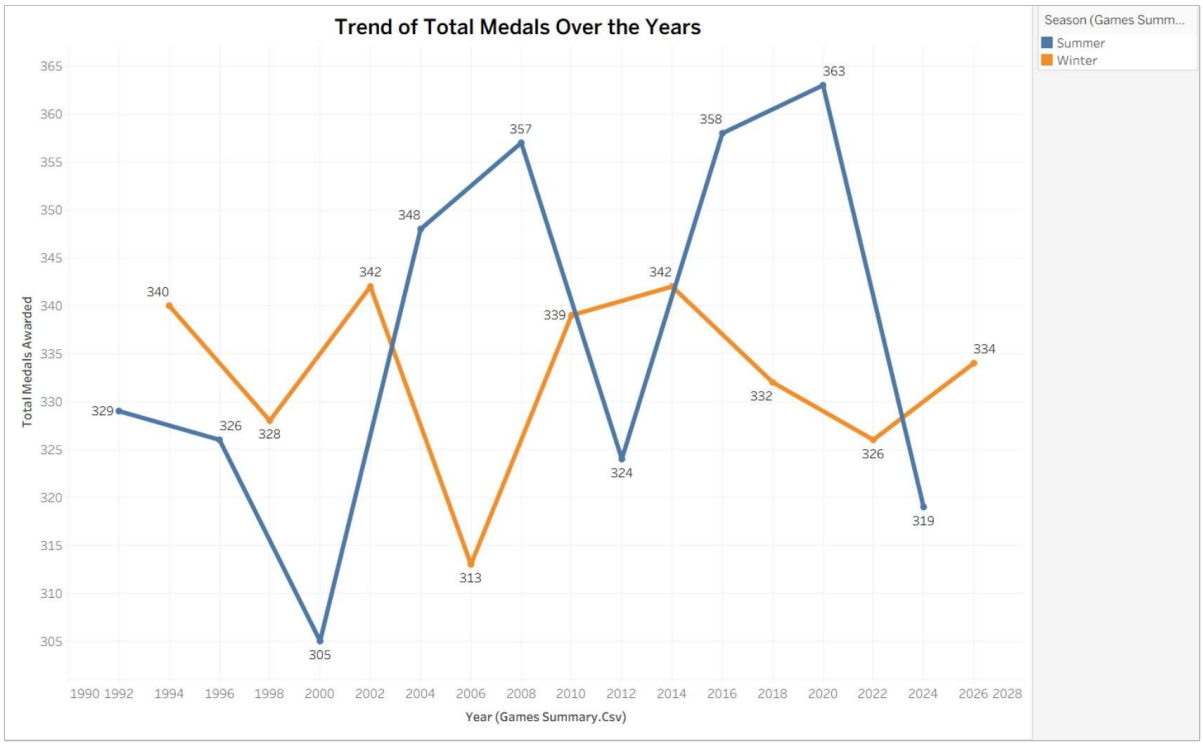


Figure 3: Trend of Total Medals Over the Years (Line Chart)

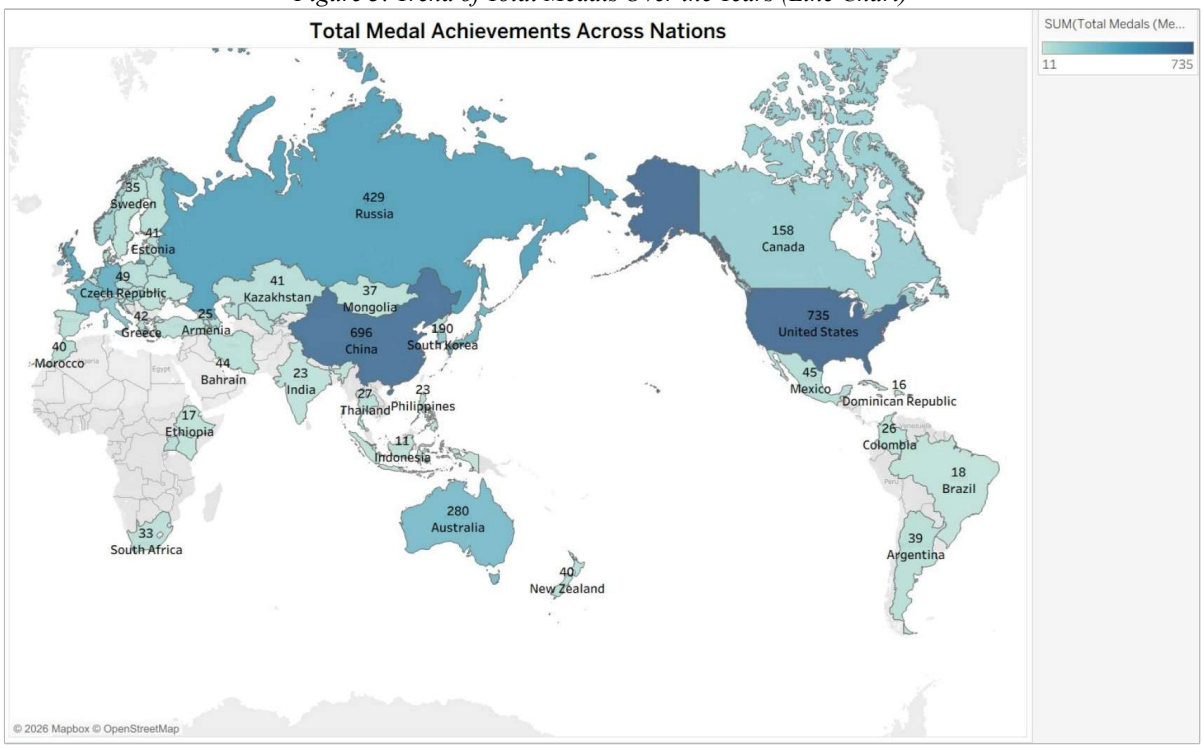


Figure 4: Total Medal Achievements Across Nations (Filled Map)

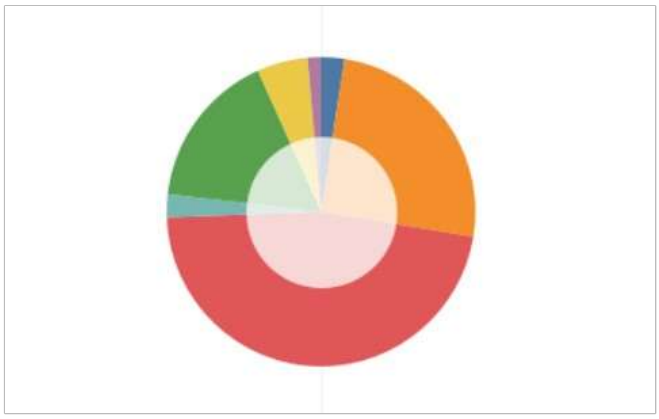


Figure 5: Medal Share by Region (Donut Chart)

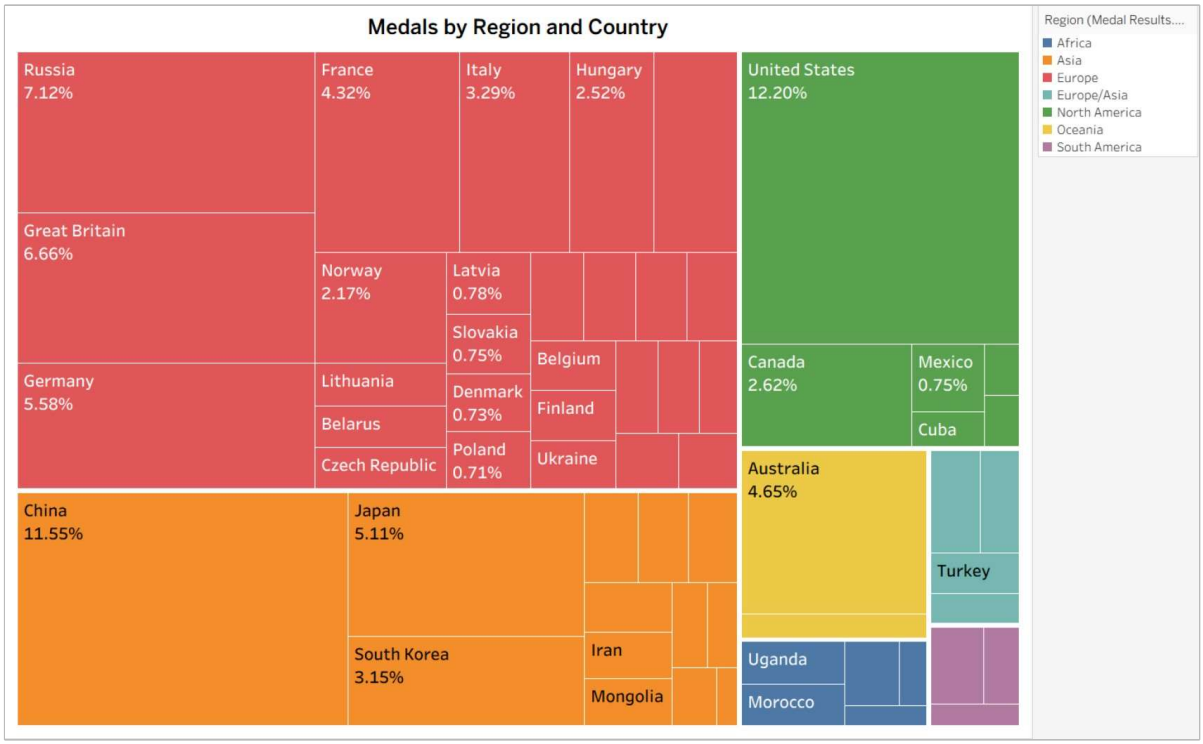


Figure 6: Medals by Region and Country (Treemap)

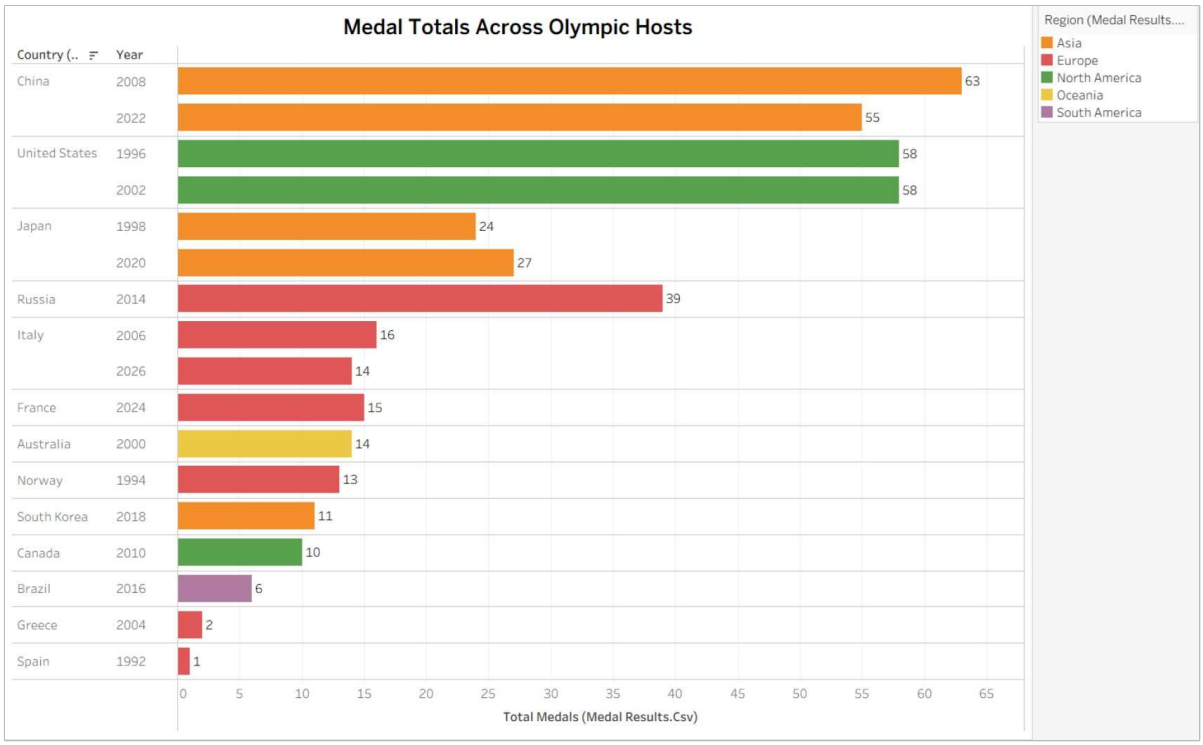


Figure 7: Medal Totals Across Olympic Hosts (Horizontal Bar)

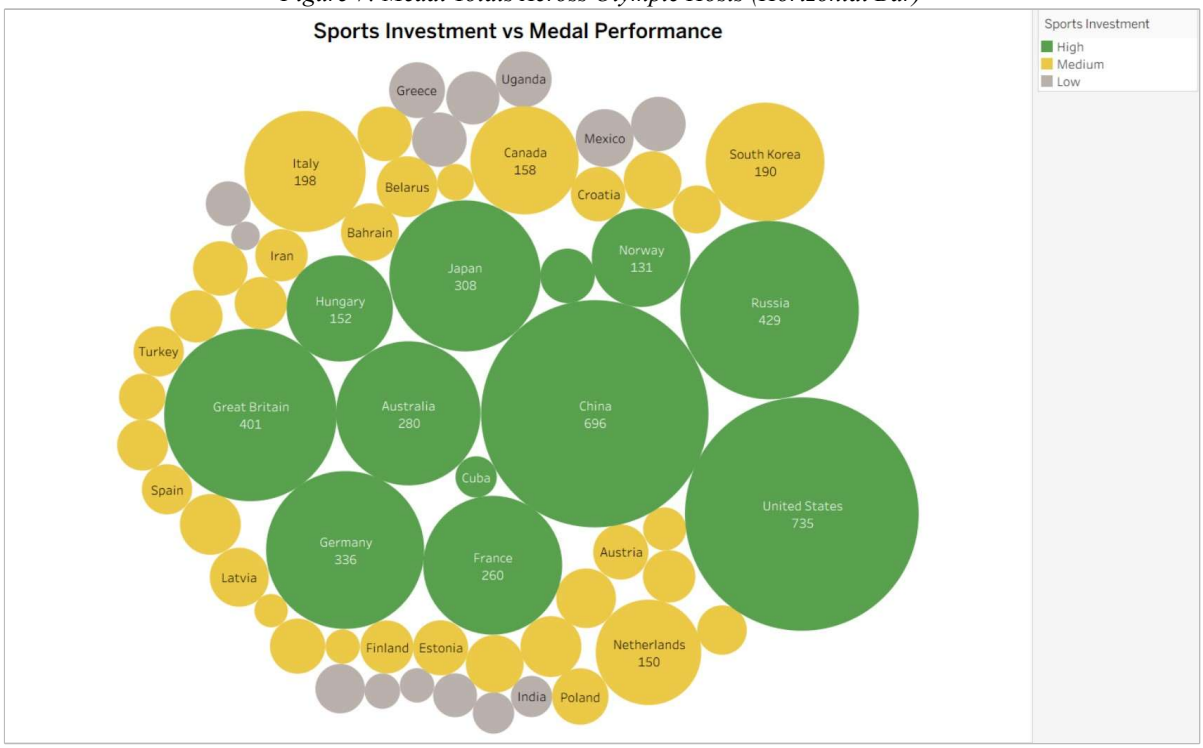


Figure 8: Sports Investment vs Medal Performance (Packed Bubbles)

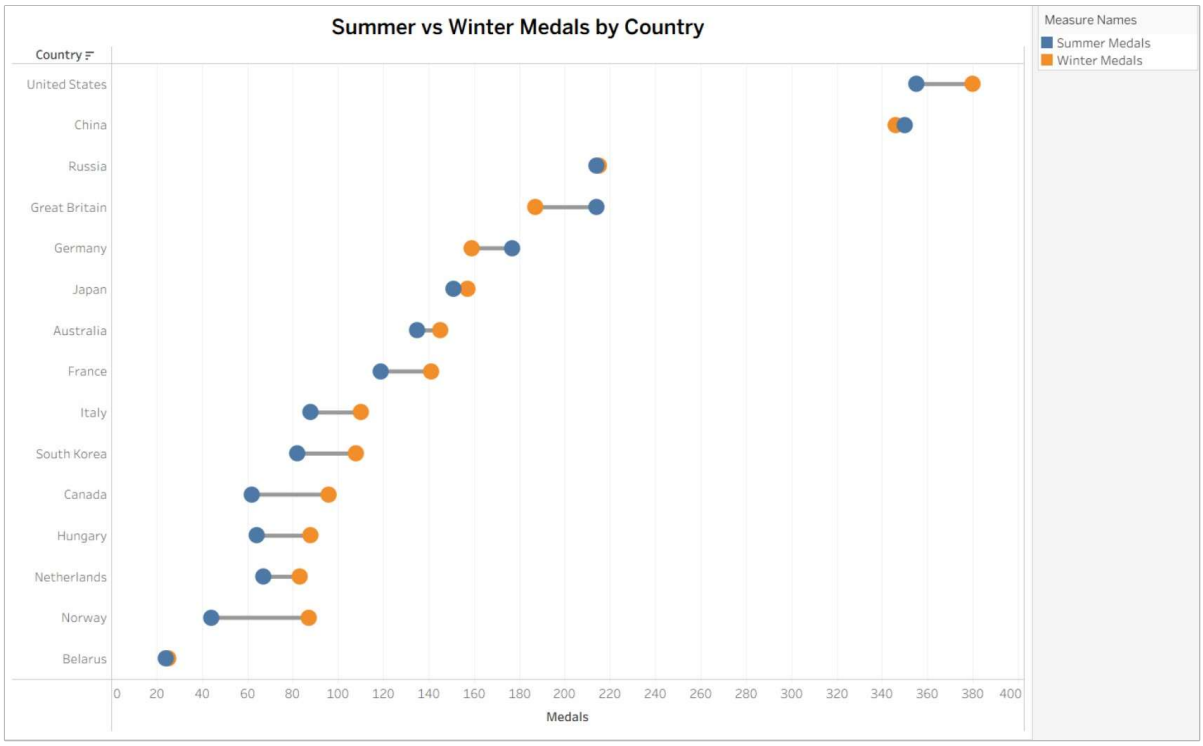


Figure 9: Summer vs Winter Medals by Country (Connected Dot Plot)

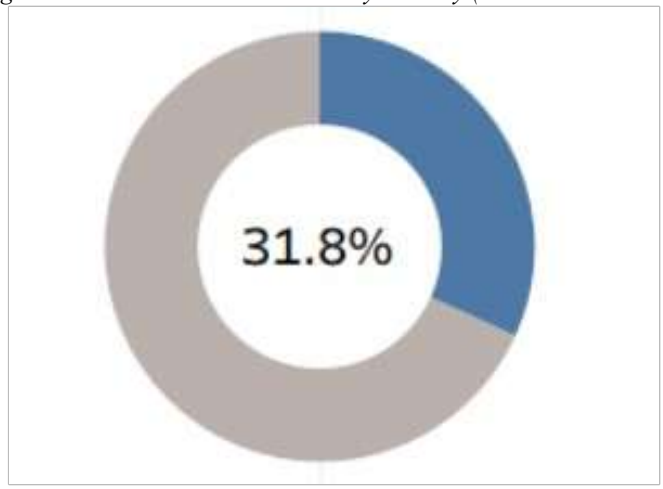


Figure 10: Average Medal Efficiency (Radial Gauge)

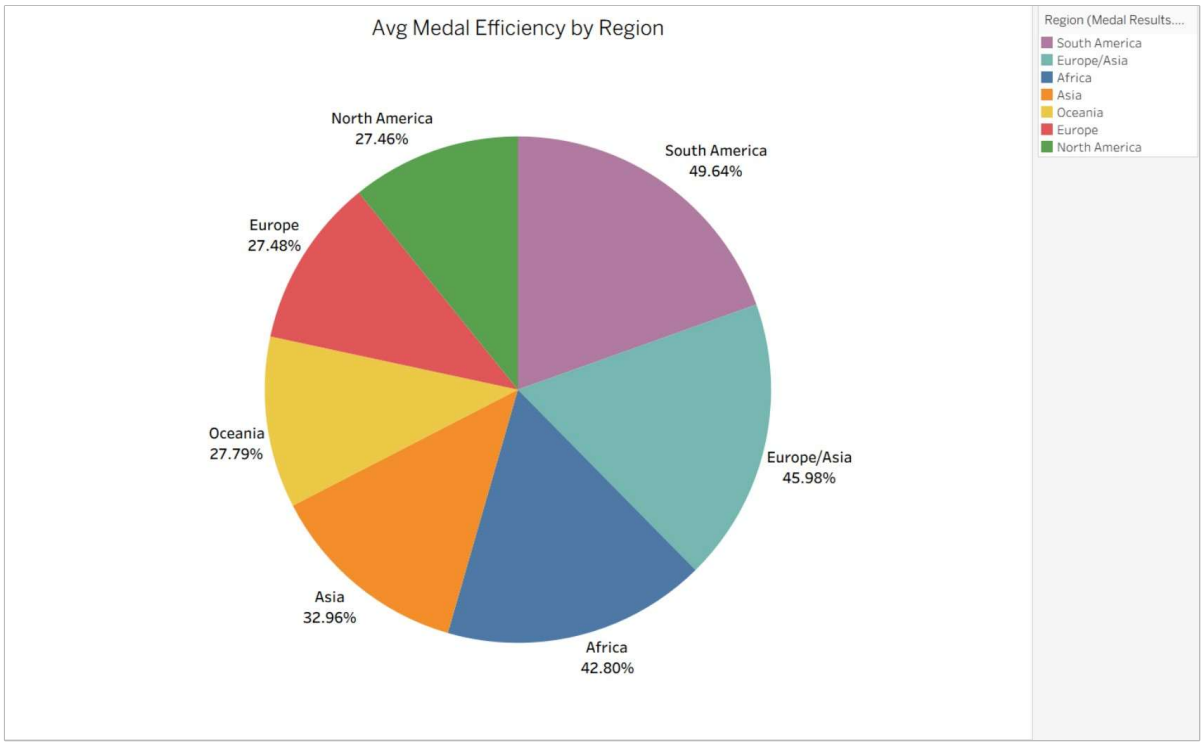


Figure 11: Avg Medal Efficiency by Region (Pie Chart)



Figure 12: KPI Cards – Total Medals, Nations, Athletes, Olympic Games

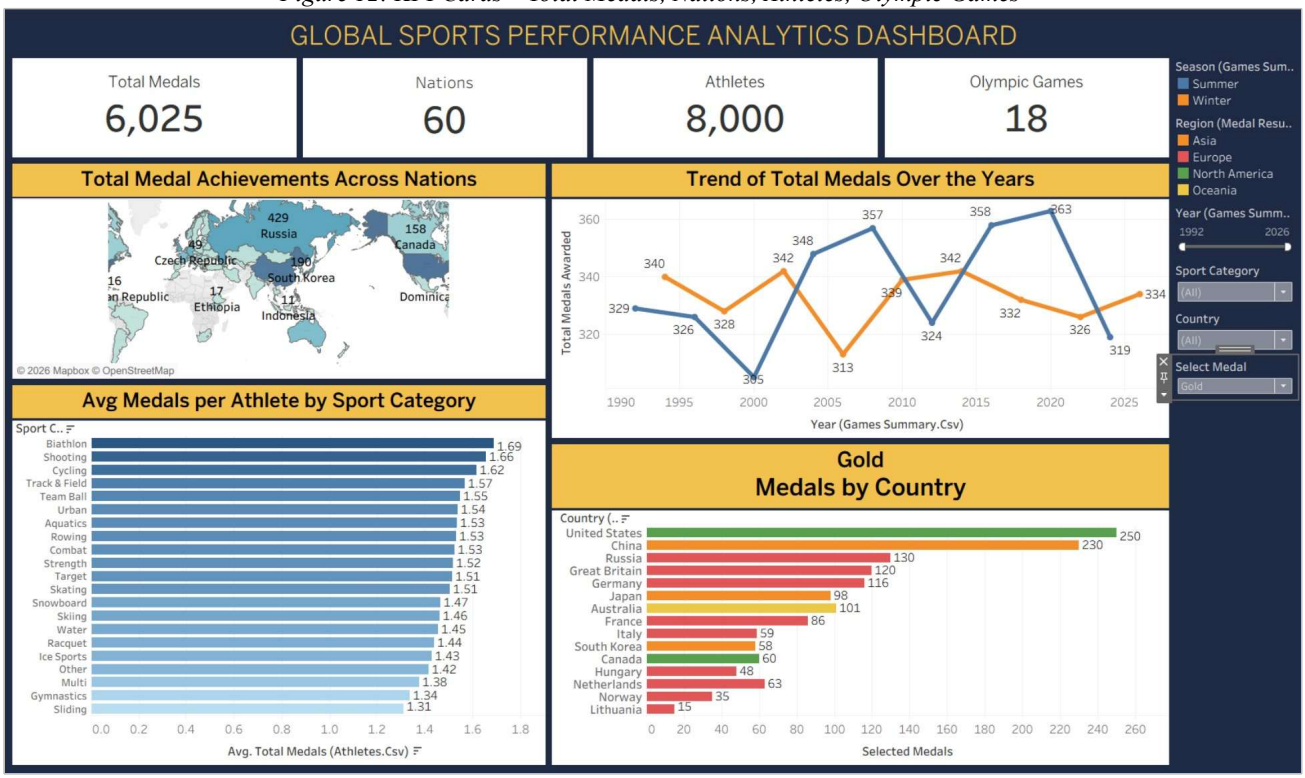


Figure 13: Global Sports Performance Analytics Dashboard (Select Medal parameter set to Gold)

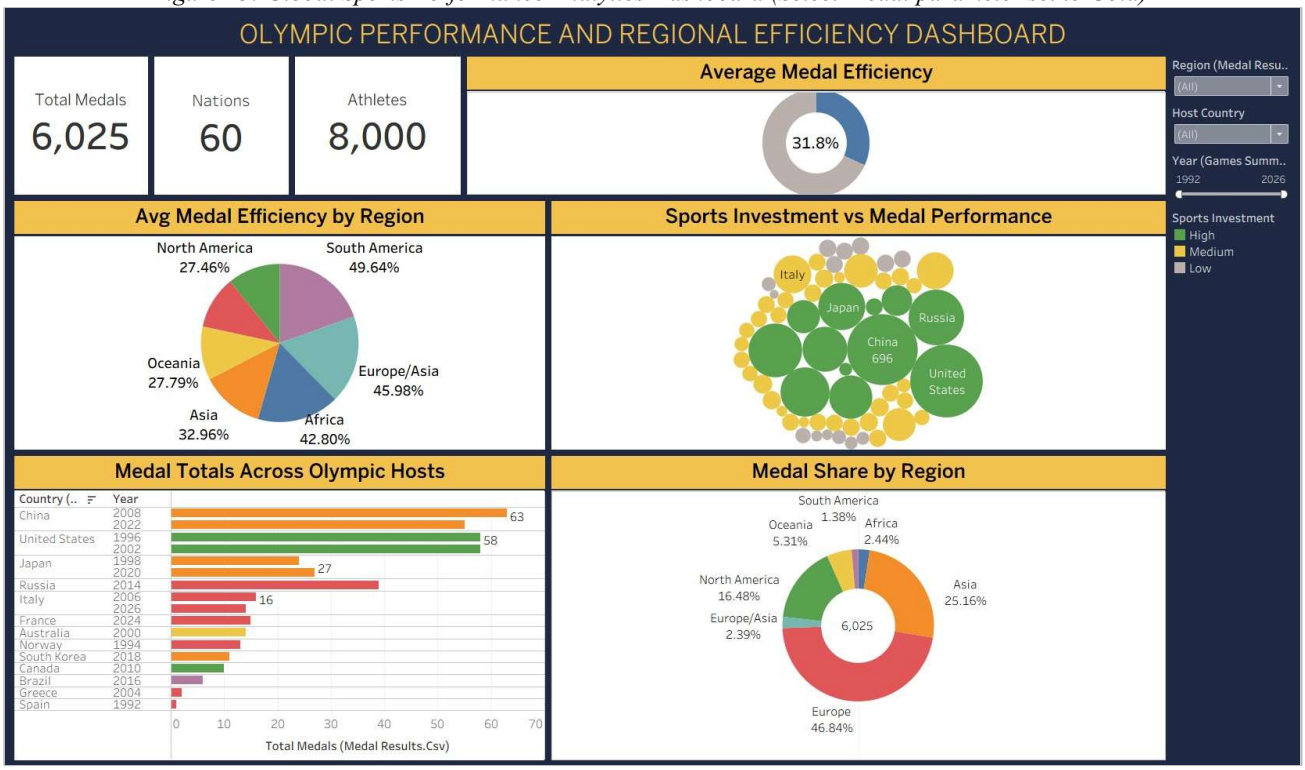
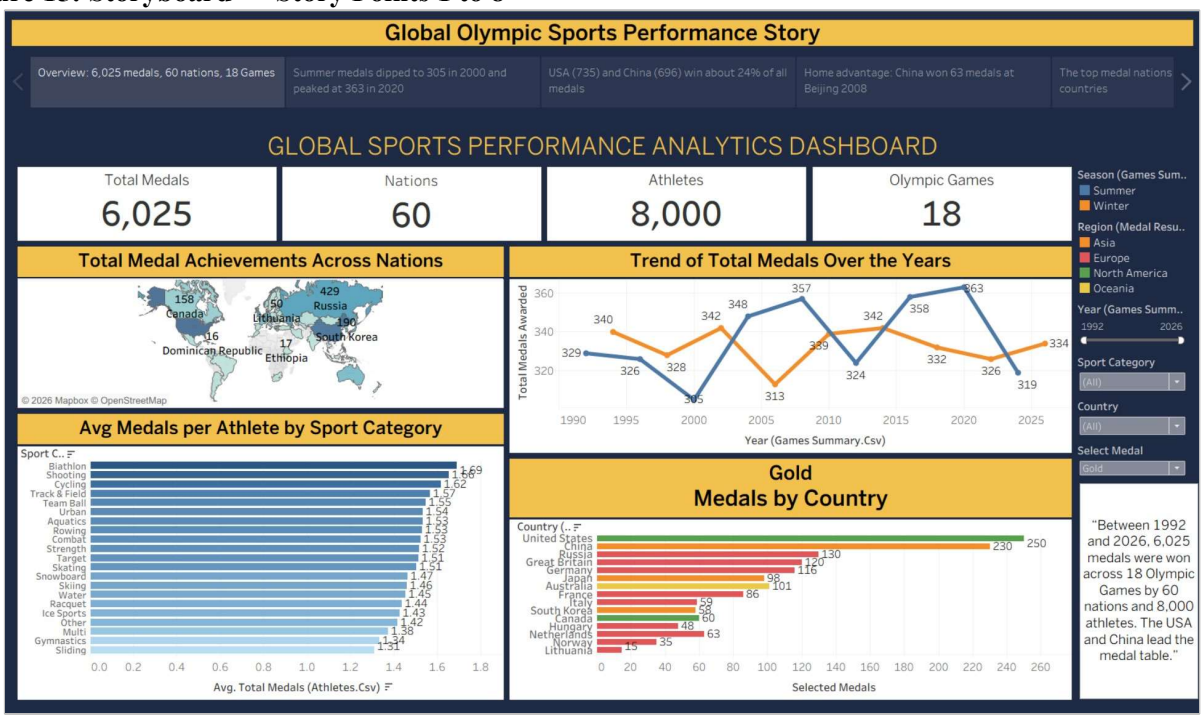
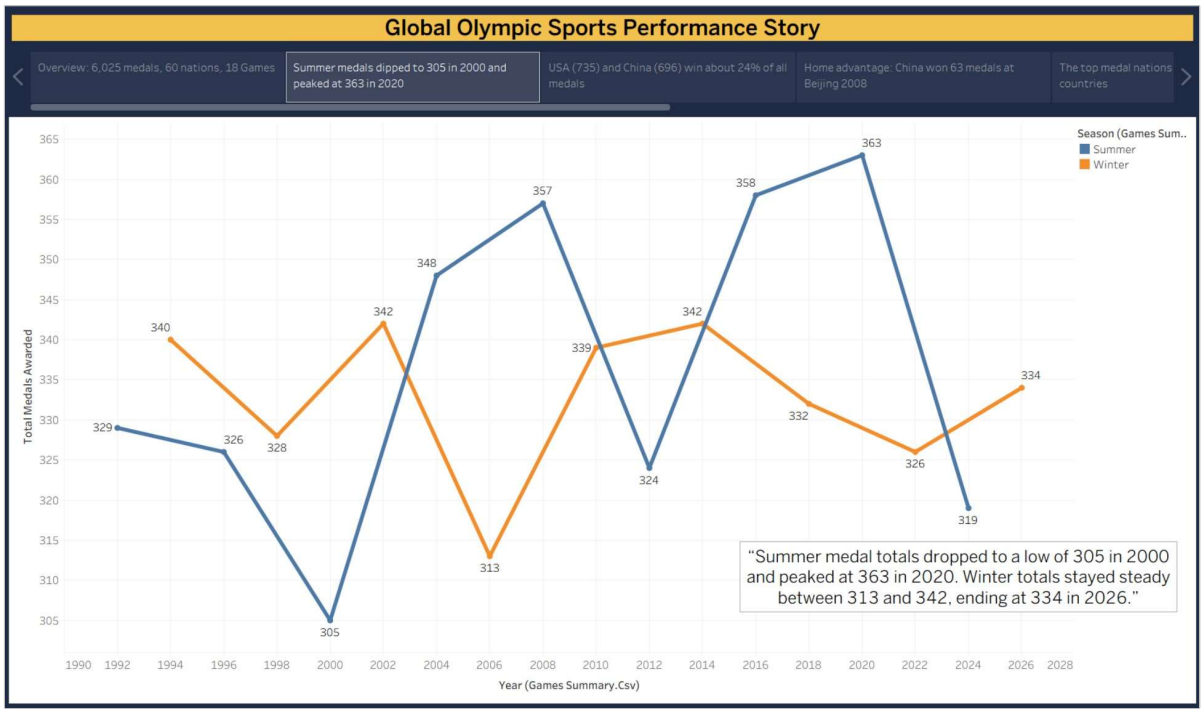


Figure 14: Olympic Performance and Regional Efficiency Dashboard

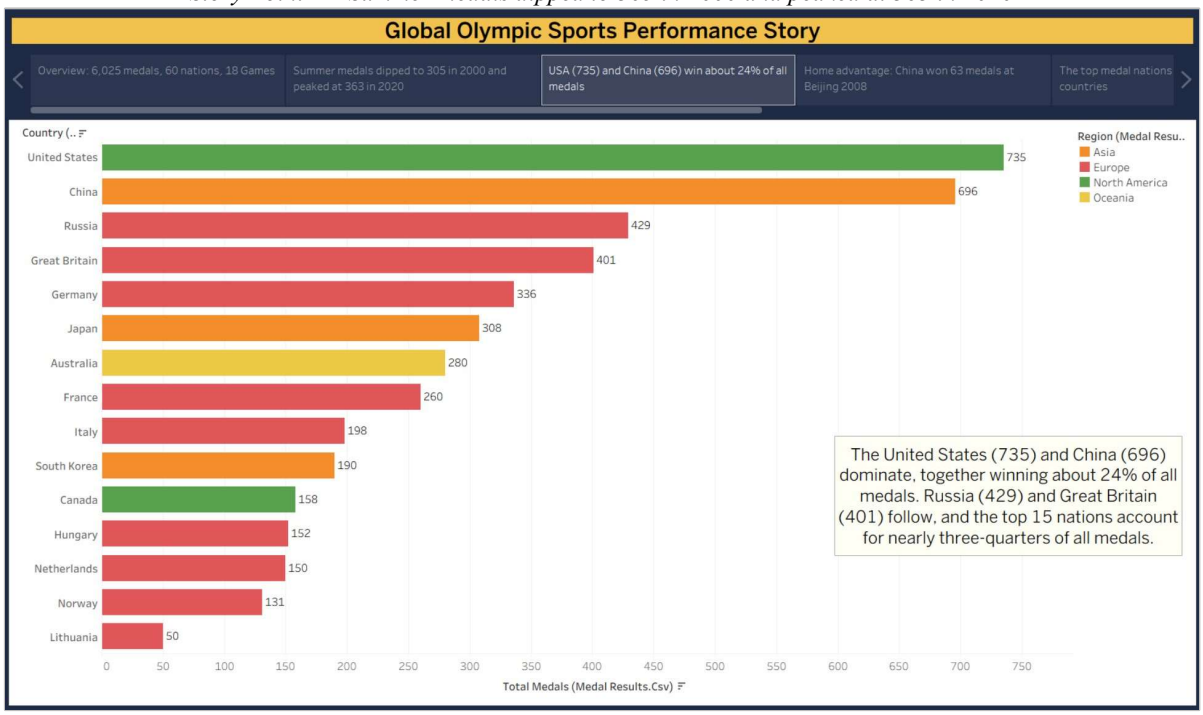
Figure 15: Storyboard — Story Points 1 to 8



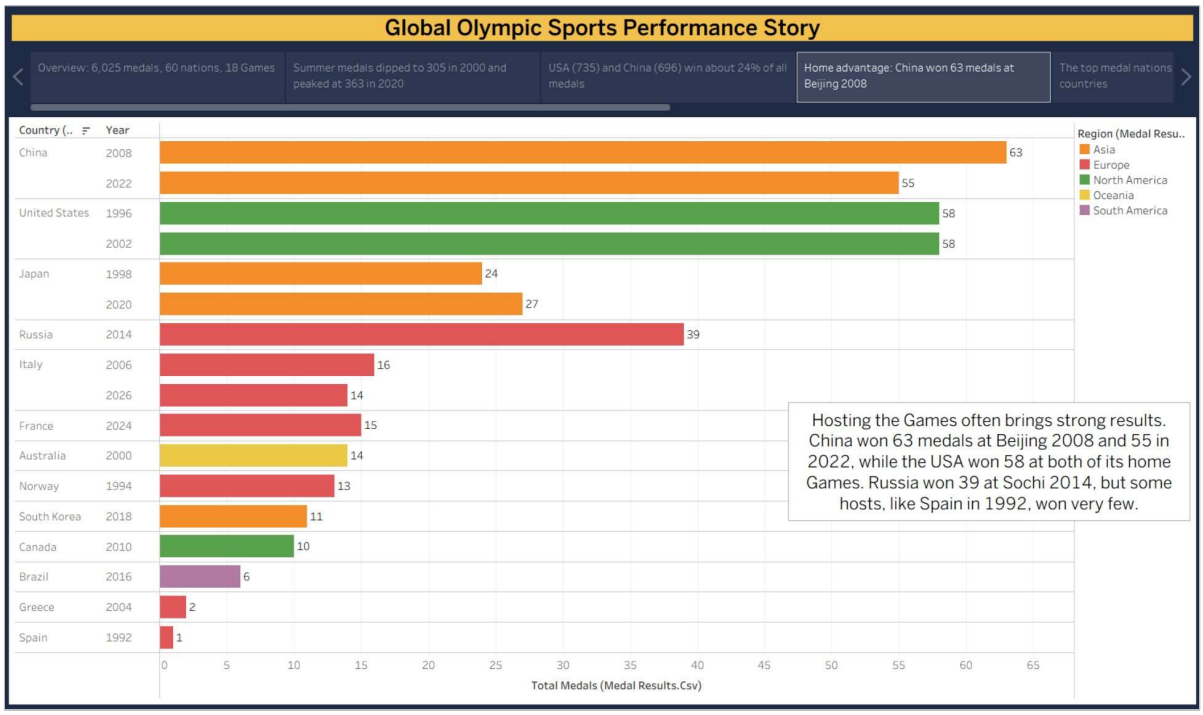
Story Point 1 – Overview: 6,025 medals, 60 nations, 18 Games



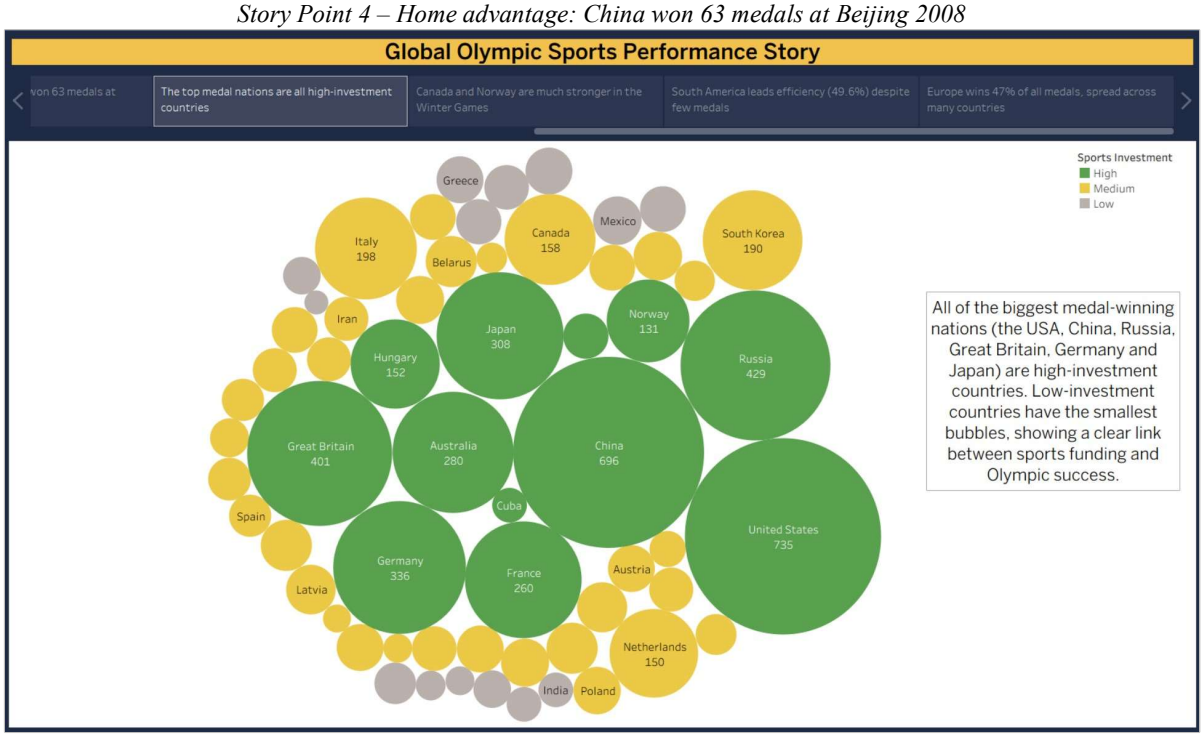
Story Point 2 – Summer medals dipped to 305 in 2000 and peaked at 363 in 2020



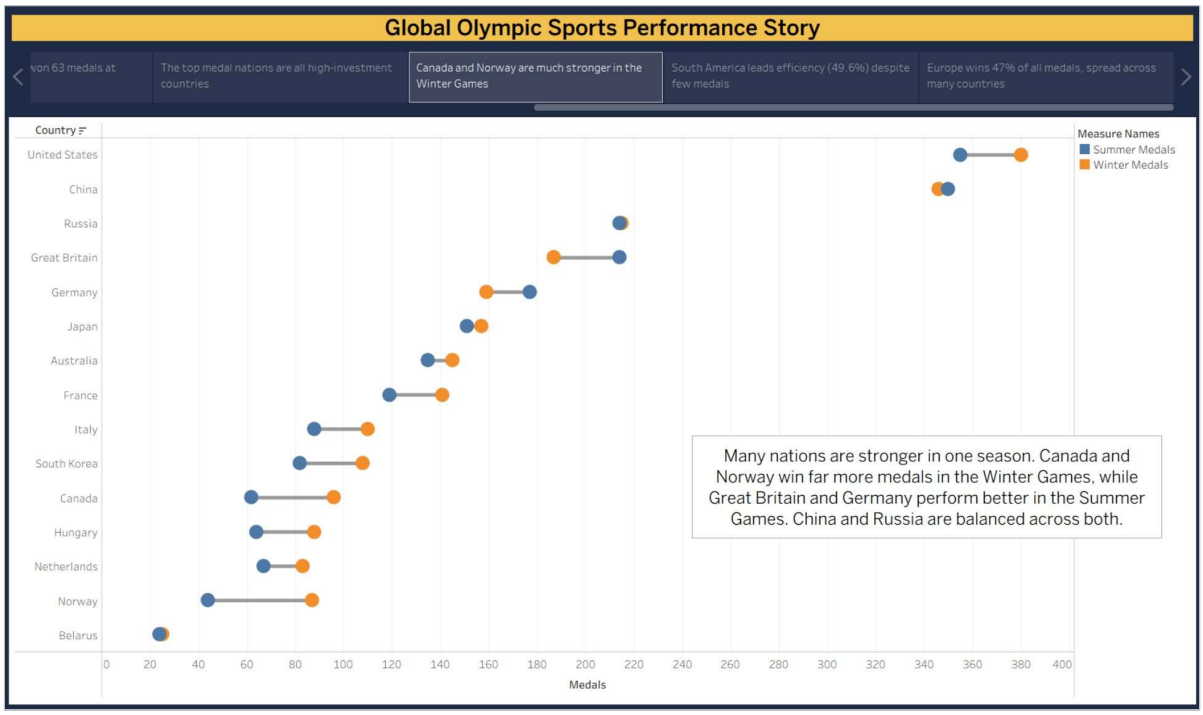
Story Point 3 – USA (735) and China (696) win about 24% of all medals



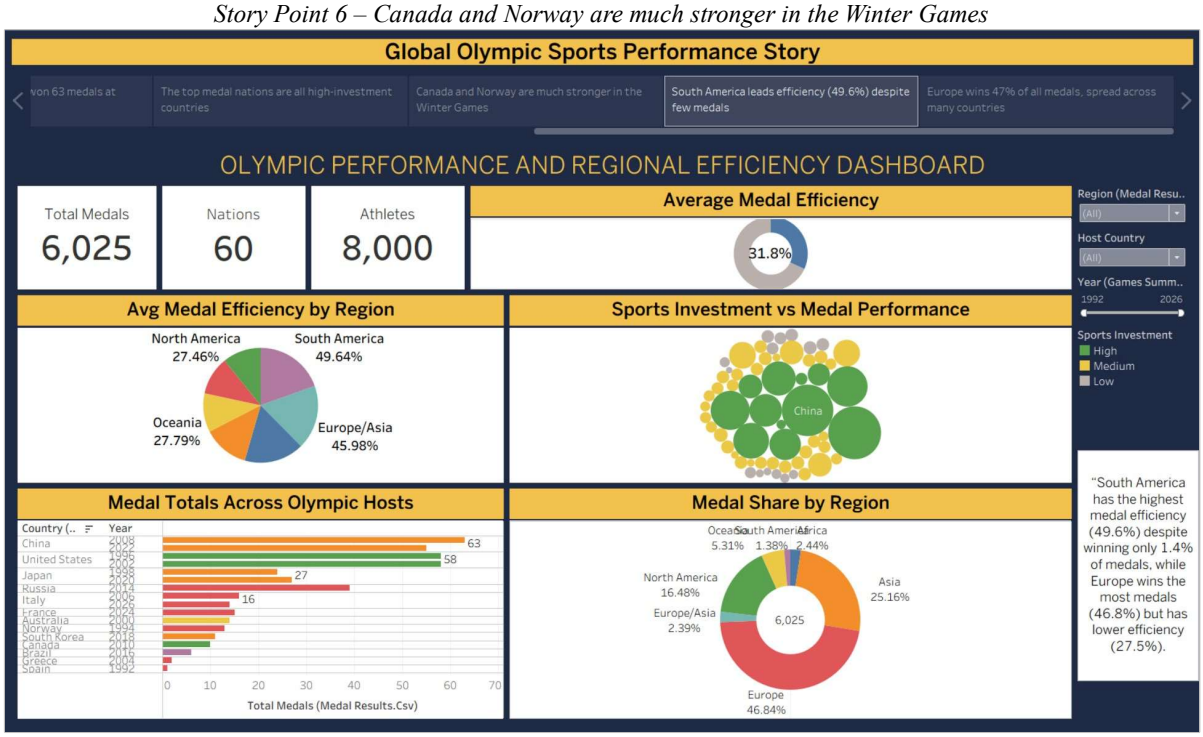
Story Point 4 – Home advantage: China won 63 medals at Beijing 2008



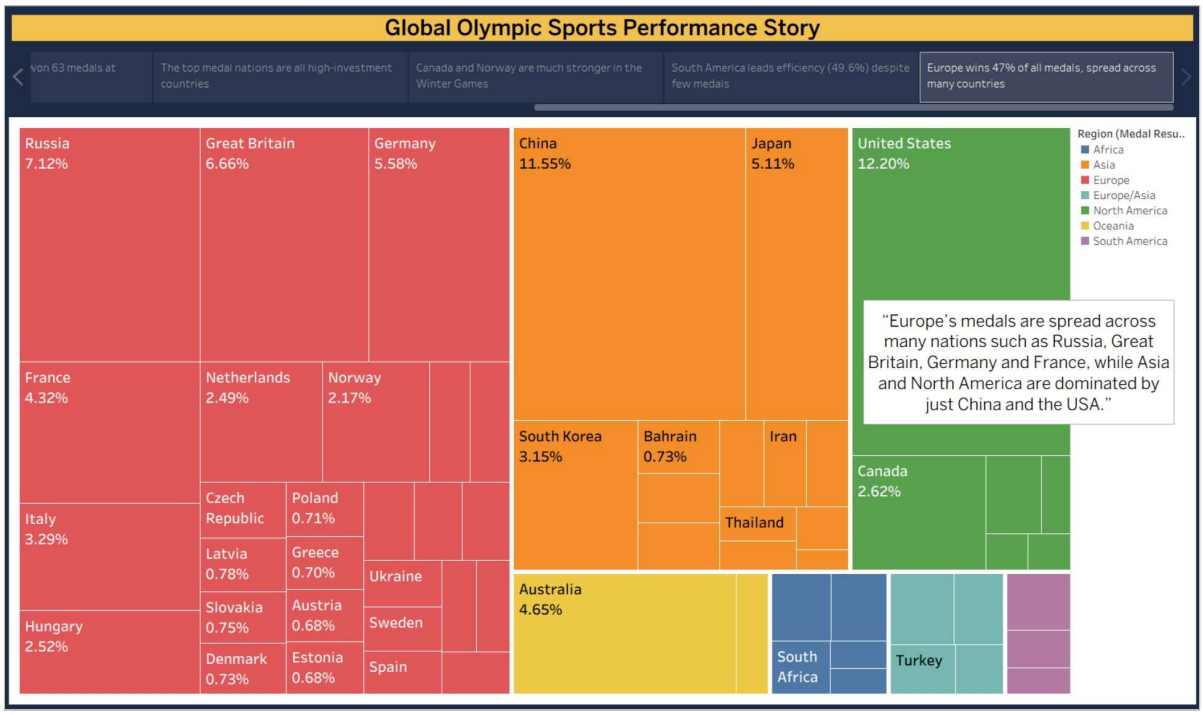
Story Point 5 – The top medal nations are all high-investment countries



Story Point 6 – Canada and Norway are much stronger in the Winter Games



Story Point 7 – South America leads efficiency (49.6%) despite few medals



Story Point 8 – Europe wins 47% of all medals, spread across many countries

Tableau Public Links:

Dashboard 1:
https://public.tableau.com/views/GlobalOlympicSportsPerformanceAnalysis/GLOBALSPORTSPERFORMANCEANALYTICSDASHBOARD

Dashboard 2:
https://public.tableau.com/views/GlobalOlympicSportsPerformanceAnalysis/OLYMPICPERFORMANCEANDREGIONALEFFICIENCYDASHBOARD

Storyboard:
https://public.tableau.com/views/GlobalOlympicSportsPerformanceAnalysis/OlympicStory

7.2 Web Integration

The published views were embedded into a Flask web application using the Tableau Embedding API v3 (<*tableau-viz*> web component). The site uses Bootstrap 5 with the same navy and gold theme as the dashboards.

Route	Page
/	Home – hero section, KPI numbers and six key-insight cards
/about	Problem statement, dataset summary, users and tools
/dashboard	Embedded Global Sports Performance Analytics Dashboard

/efficiency	Embedded Olympic Performance & Regional Efficiency Dashboard
/story	Embedded 8-scene Tableau story

Project structure:

```
olympic_flask/
├── app.py              (Flask routes + Tableau view URLs)
├── requirements.txt
├── templates/          (base, index, about, viz)
└── static/css/style.css
```

Run with: `pip install flask` then `python app.py` and open http://127.0.0.1:5000. All five routes were tested and returned HTTP 200.

8. Advantages & Disadvantages

Advantages

- Consolidates four related files (~9,000 rows) into two easy-to-navigate dashboards and a story.
- Combines medal counts with efficiency, investment and host status in one view instead of isolated
- reports. Interactive filters and the Select Medal parameter let users explore any year range, region, country, sport category or medal type.
- The eight-scene story makes insights accessible to non-technical viewers.
- Published free on Tableau Public and embedded in a Flask website, accessible from any browser.
- Modular workbook and Flask routes allow new sheets, Games or pages to be added easily.

Disadvantages

- The dataset is a static CSV snapshot and does not update automatically.
- Investment levels are categorical (High / Medium / Low), so exact spending cannot be analysed.
- Tableau Public views are public, so the approach is not suited to confidential data.
- The Flask site runs on a local development server and is not yet deployed to the cloud.
- Very small slices (e.g. South America and Africa in the donut) are hard to label and rely on tooltips.

9. Conclusion

This project turned the Olympic Games Performance & Medal History dataset (1992–2026) into an interactive Tableau solution that explains what drives Olympic success. The analysis shows that the United States (735) and China (696) dominate the medal table, host nations perform strongly at home (China won 63 medals at Beijing 2008), every top medal nation is a high-investment country, and smaller regions can be highly efficient (South America leads at 49.6%). Through relationships across four files, seven calculated fields, a parameter, twelve visualizations, two responsive dashboards, an eight-scene story and a Flask web integration, the project gives Olympic committees, investors and journalists a clear, shareable decision-support tool. Both sprints were delivered on schedule.

10. Future Scope

- Connect to a live or regularly refreshed data source after each Olympic Games.
- Add predictive models to forecast medal counts for upcoming Games.
- Include exact investment amounts, GDP and population for deeper efficiency analysis.
- Deploy the Flask website to a cloud platform (e.g. Render or PythonAnywhere) with a public URL.
- Add athlete-level drill-downs (gender, age, training hours) and a "how to read this dashboard" guide.

11. Appendix

Source Code

The Tableau workbook (Global_Olympic_Sports_Performance_Analysis.twbx) and the Flask web application source code (folder *olympic_flask*) are available in the GitHub repository. **Dataset Link**

https://www.kaggle.com/datasets/meruvakodandasuraj/olympic-games-performance-and-medal-history-199226

GitHub & Project Demo Link

GitHub Repository Link:
https://github.com/ManasGeedkar/Global-Olympic-Sports-Performance-Analysis

Tableau Public Link:
https://public.tableau.com/app/profile/manas.geedkar/vizzes

Project Demonstration Link:
https://drive.google.com/file/d/16cYlP0-HJqdKLURIMIJ88-U61SnViFC7/view?usp=sharing